Bank Customer Attrition Prediction

Executive Summary

This report presents a comprehensive analysis of customer attrition at a bank, leveraging a dataset of 10,000 customers. The core objective was to develop a predictive model capable of accurately identifying customers at high risk of churn, facilitating proactive interventions and enhancing retention strategies. The analysis encompassed exploratory data analysis (EDA), rigorous feature engineering, model selection, and thorough performance evaluation. Key findings revealed that credit score, age, account balance, customer complaints, satisfaction score, and activity status are significant predictors of churn. The Support Vector Machine (SVM) emerged as the preferred model, demonstrating a strong balance of precision and recall, essential for minimizing false positives in churn prediction. This report details the process, findings, and recommendations for deploying the SVM model to effectively reduce customer attrition.[3]

Problem Statement

Customer attrition poses a substantial threat to the financial health and long-term sustainability of banking institutions. The costs associated with acquiring new customers far exceed those of retaining existing ones. Addressing customer attrition requires a proactive, data-driven approach that enables the bank to identify and engage at-risk customers with timely and effective retention strategies. By understanding the key drivers of churn and predicting customer behavior, the bank can enhance customer satisfaction, loyalty, and overall financial performance. The central problem addressed by this project is the development of a predictive model to identify customers likely to churn, enabling proactive interventions.

Data Description

The "Bank Customer Attrition Insights" dataset comprises 10,000 customer records, each described by 18 attributes. These attributes include:

- Demographic Information: CreditScore, Age, Geography, Gender
- Account Details: Balance, Tenure, NumOfProducts, HasCrCard, IsActiveMember, EstimatedSalary
- Customer Behavior: Complain, Satisfaction Score, Point Earned, Card Type
- Identifiers: RowNumber, CustomerId, Surname
- Target Variable: Exited (Binary: 0 = Retained, 1 = Churned)

The dataset provides a comprehensive view of customer profiles, enabling a detailed analysis of factors contributing to attrition. A critical challenge is the class imbalance, with only 20.4% of customers having churned, requiring careful consideration during model development and evaluation.

Exploratory Data Analysis

The EDA phase aimed to uncover patterns, relationships, and anomalies within the dataset. Key findings from the EDA include:

- Credit Score and Churn: Customers who churned tend to have lower credit scores. The distribution of credit scores for churned customers is skewed towards lower values, indicating a potential correlation. [4]

- Age and Balance: Churned customers are distributed across various ages and balances but are concentrated more among those with lower balances. This suggests that customers with lower account balances may be more likely to leave.[4]

- Correlation Analysis: A heatmap of feature correlations revealed significant relationships between the 'Exited' status and 'Balance' and 'Satisfaction Score.' This highlights the importance of financial factors and customer satisfaction in predicting churn.[4]

The EDA also identified potential issues, such as outliers in numerical features and the need for feature transformations to improve model performance. The insights gained from the EDA informed subsequent feature engineering and model selection steps.

Feature Engineering

Feature engineering involved transforming and selecting relevant features to enhance the predictive power of the models. The following steps were undertaken:

Feature Selection: Irrelevant features such as RowNumber, CustomerId, and Surname were excluded from the analysis.

Encoding Categorical Variables:

- Geography and Card Type were one-hot encoded to convert categorical data into numerical format.
- Gender was binary encoded.
- The satisfaction Score was ordinally encoded to preserve the order of satisfaction levels.

Numerical Feature Transformation:

- Balance was log-transformed to address skewness and improve normality.
- EstimatedSalary was robustly scaled to mitigate the impact of outliers.

Feature Scaling: CreditScore and Point Earned were scaled using standard and min-max scaling, respectively, to ensure consistent ranges across features.

These feature engineering steps aimed to create a dataset suitable for machine learning models, improving their accuracy and interpretability.

Modeling Results

Data Partitioning Strategy

Several machine learning models were trained and evaluated, including Logistic Regression, Decision Trees, K-Nearest Neighbors, Support Vector Machines (SVM), Random Forest, XGBoost, Gradient Boosting, AdaBoost, and LightGBM. The dataset was partitioned into training (60%), validation (20%), and test (20%) sets using stratified sampling to maintain class balance.[1]

- Model Selection: Based on initial results, the models were performing really well, as they were overfitting over the data. The whole analysis was dependent on a single variable, thus, it was not a generalized model. The initial model results were:

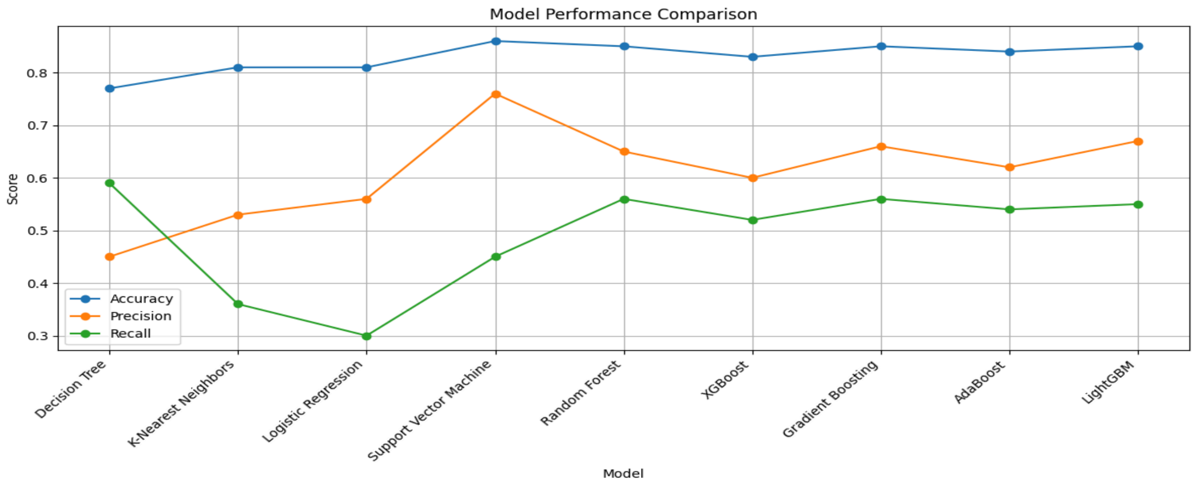
Model	Accuracy	Precision	Recall
Logistic Regression	0.85	0.71	0.47
K-Nearest Neighbors	0.75	0.21	0.08
Decision Tree	1.00	1.00	0.98
Support Vector Machine	0.80	0.00	0.00
Random Forest	1.00	1.00	1.00
XGBoost	1.00	1.00	1.00
Gradient Boosting	1.00	1.00	1.00
AdaBoost	1.00	1.00	1.00
LightGBM	1.00	1.00	1.00

Feature Importance Analysis: A Random Forest Classifier feature importance plot was used to assess which features most influence the outcome. Features like "Credit Score," "Age," and "Balance" were among the highest. Low-importance features are considered for removal to reduce model complexity. The complain variable was also dropped as it was the reason for the overfitting of the model.

After Feature Importance, since Complain seemed to be highly relevant and was the reason for overfitting, after removing it, the model's performance is as follows:

Model	Accuracy	Precision	Recall
Decision Tree	0.77	0.45	0.59
K-Nearest Neighbors	0.81	0.53	0.36
Logistic Regression	0.81	0.56	0.30
Support Vector Machine	0.86	0.76	0.45
Random Forest	0.85	0.65	0.56
XGBoost	0.83	0.60	0.52
Gradient Boosting	0.85	0.66	0.56
AdaBoost	0.84	0.62	0.54
LightGBM	0.85	0.67	0.55

Several models were trained and Support Vector Machine (SVM) emerged as the preferred model due to its high accuracy and precision. The higher precision indicates a lower false positive rate, which is desirable in this context, and the overall results indicate a strong and balanced performance across all evaluation metrics.



Performance Metrics
Support Vector Machine
Accuracy: 0.86
Precision: 0.76
Recall: 0.45

Deployment Recommendations

- Pipeline: Implement a real-time prediction pipeline with data ingestion, feature engineering (encoding, scaling, transformation), and SVM scoring to generate churn probabilities.
- Interventions: Develop personalized retention strategies based on SVM predictions and feature importance (CreditScore, Geography, Age, Balance, Satisfaction Score).
- Monitoring: Continuously monitor model performance (AUC-ROC, F1-Score, confusion matrix) and retrain regularly.
- A/B Testing: Use A/B testing to evaluate the effectiveness of different retention strategies.
- CRM Integration: Integrate the model with the CRM system for targeted customer interactions.
- Class Imbalance: Address the class imbalance (20.4% churn) using techniques like oversampling or undersampling.

Anticipated Challenges and Solutions

- Feature Engineering: Employ feature engineering techniques such as encoding for categorical variables, and PCA for dimensionality reduction.
- Outliers and Missing Data: EDA techniques to detect and treat outliers.
- Interpretability vs. Performance: Begin with less complex models and apply model-agnostic interpretation techniques for complex models.

Business Implications

The successful deployment of the attrition prediction model offers several significant business benefits:

- Improved Customer Retention: Reduce churn rates by proactively identifying and engaging at-risk customers.
- Cost Reduction: Achieve cost savings in customer acquisition and onboarding by retaining existing customers.
- Enhanced Customer Experience: Address pain points in the customer journey, leading to improved customer satisfaction.
- Increased Revenue: Drive revenue through retained customers and their continued engagement with banking products and services.
- Competitive Advantage: A lower churn rate can be a key differentiator in the competitive banking industry.

Conclusion

This project successfully developed an SVM model for predicting customer attrition at a bank. The model demonstrates strong performance and the potential to significantly improve customer retention efforts. By integrating the model into the bank's operations, the bank can enhance customer satisfaction, reduce costs, and gain a competitive advantage.

References

[1] Kohavi, R. (1995). A study of cross-validation and bootstrap for accuracy estimation and model selection. International Joint Conference on Artificial Intelligence.

[2] Provost, F., & Fawcett, T. (2013). Data Science for Business: What you need to know about data mining and data-analytic thinking. O'Reilly Media.

[3] Reichheld, F. F., & Teal, T. (1996). The Loyalty Effect: The Hidden Force Behind Growth, Profits, and Lasting Value. Harvard Business School Press.

[4] Earlier Assignment Reports- Can be found here

[5] The code for this project can be found here